

Facial Expression Recognition

Submission Report

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Summary

This report documents the full development process for a 7-class facial emotion classifier, trained on a dataset of 48×48 grayscale face crops (28,709 train / 7,178 test images). The approach evolved through eight iterations, moving from a single fine-tuned ResNet18 baseline to a final heterogeneous ensemble.

The final submitted model is a three-model heterogeneous ensemble (ResNet34 + EfficientNet-B0 + ResNet18) trained with a WeightedRandomSampler to directly oversample the severely underrepresented *disgust* class, combined with strong augmentation, Mixup, label smoothing, and test-time augmentation (TTA). This achieved 0.707 categorical accuracy on the public leaderboard. A second submission - a two-model heterogeneous ensemble (ResNet34 + EfficientNet-B0) without the sampler - scored 0.71301 on the public split, ranked #33.

1. Dataset and Exploratory Data Analysis

- Source: train/ (28,709 images) and test/ (7,178 images), 48×48 grayscale face crops, plus train.csv (id → label) and test.csv (id to predict).
- 7 target classes: angry, disgust, fear, happy, neutral, sad, surprise.
- Validation split for the final two runs: stratified train_test_split, test_size=0.15, random_state=42 → Train 24,402 / Val 4,307. (Earlier iterations used a 90/10 split — noted per-iteration in Section 2.)

Full training-set class distribution:

Class	Count	Share of train set
happy	7,215	25.1%
neutral	4,965	17.3%
sad	4,830	16.8%
fear	4,097	14.3%
angry	3,995	13.9%
surprise	3,171	11.0%
disgust	436	1.5%

The disgust class is the central challenge in this dataset: it has roughly 16× fewer examples than happy. A model optimizing plain accuracy has little incentive to learn it well, since misclassifying it barely moves the overall score. This motivated class-weighted loss in every iteration, and direct oversampling via WeightedRandomSampler in the final one.

2. Approach Evolution

#	Approach	Key techniques	Result
0	Code check	Random label assignment, no model trained	Public LB: 0.138 (matches theoretical random-chance floor of $1/7 \approx 0.143$)
1	ResNet18 transfer learning	ImageNet-pretrained ResNet18, replaced final layer, class-weighted cross-entropy, basic augmentation (flip/rotate/jitter), 10 epochs, Adam	Public LB: 0.658. Validation accuracy peaked at epoch 9 (0.671) then fell at epoch 10 (0.665) — the submission used the last epoch, not the best one.
2	+ best-checkpoint saving, longer training, TTA	25 epochs, CosineAnnealingLR, save only the best-validation checkpoint (not the last epoch), test-time augmentation (average of normal + flipped prediction)	Public LB: 0.699. Best val_acc 0.676.
3	+ Dropout, weight decay	Added Dropout(0.3) and weight_decay=1e-4 to counter an observed overfitting pattern (train loss near zero while val accuracy plateaued)	Val_acc improved to 0.681 (fair like-for-like comparison), but public score came back at 0.690 — attributed to sampling noise on the small public split rather than a genuine regression.

#	Approach	Key techniques	Result
4	ResNet18 + ResNet34 same-family ensemble (attempt)	Two-model soft-vote ensemble with TTA	Training session disconnected mid-run ($\approx 52\%$ through); no result obtained. Directly motivated the batch-level heartbeat logging and checkpoint-first design used in later scripts.
5	EfficientNet-B0 single model (attempt)	Swapped backbone, reused ResNet18's exact learning-rate and epoch settings unchanged	Public LB: 0.642. Diagnosis: train_acc reached only 0.767 (vs. ResNet18's ≈ 0.98) - the model was underfitting, not overfitting. Confirmed that swapping architectures requires re-tuning the training recipe, not just the backbone.
6	ResNet18 dual-seed ensemble (fallback)	Same proven ResNet18 recipe, two random seeds, soft-vote + TTA	Prepared as a low-risk fallback; superseded by the stronger heterogeneous-architecture strategy below before a separate result was recorded.
7	Heterogeneous ensemble: ResNet34 + EfficientNet-B0	224px input, RandAugment + RandomResizedCrop + ColorJitter + RandomErasing, Mixup ($\alpha=0.2$), label smoothing (0.1), AdamW + CosineAnnealingLR + AMP, validation-accuracy-weighted soft-vote + TTA	Best val: ResNet34 0.704 (epoch 20), EfficientNet-B0 0.696 (epoch 16). Public LB: 0.71301.
8	3-model heterogeneous ensemble + WeightedRandomSampler (FINAL)	Adds ResNet18 as a third ensemble member and a WeightedRandomSampler to directly oversample rare classes, on top of iteration 7's full recipe	Best val: ResNet34 0.684, EffNet-B0 0.684, ResNet18 0.683 (equal-weighted soft-vote + TTA). Public LB: 0.707.

3. Final Methodology (Iteration 8)

3.1 Preprocessing

- Each 48×48 grayscale image is loaded, converted to RGB by replicating the single channel three times (pretrained ImageNet backbones expect 3-channel input).
- Resized to 224×224 (up from 48×48) to preserve enough facial detail : eyes, mouth, brow position - for the pretrained filters to work with.
- Normalized using true ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]), matching what the pretrained weights were originally calibrated on, rather than a simplified 0.5/0.5/0.5 normalization used in early iterations.

3.2 Model architectures

ResNet34 : a 34-layer convolutional network using residual (“skip”) connections, which let gradients flow through many layers during training without vanishing. Fine-tuned end-to-end (no frozen layers) from ImageNet-pretrained weights, with Dropout(0.4) added before the final classification layer.

EfficientNet-B0 : a different architecture family built from MBConv blocks with squeeze-and-excitation attention, which learns to weight different channels/regions of the image adaptively. Loaded via the timm library, pretrained on ImageNet, fine-tuned with Dropout(0.3).

ResNet18 : a shallower cousin of ResNet34, added as the third ensemble member specifically for its architectural diversity relative to the other two, at low additional training cost.

Rationale for heterogeneity: models from different architecture families tend to make different mistakes on the same input, since they process the image through structurally different mechanisms (plain residual blocks vs. MBConv+attention). Averaging their predictions cancels out a meaningful share of each model's individual errors - empirically, this heterogeneous combination outperformed a same-architecture, different-seed ensemble tried earlier (iteration 6).

3.3 Handling class imbalance

- Class-weighted cross-entropy loss: each class's contribution to the loss is scaled inversely to its frequency, so misclassifying a rare class (disgust) is penalized more heavily than misclassifying a common one (happy).
- WeightedRandomSampler: in addition to the loss weighting, the DataLoader itself samples rare-class images more frequently each epoch, so the model is physically shown disgust/fear examples far more often than their raw counts would otherwise allow.
- Label smoothing (0.1): softens the training target from “100% confident” to “ $\sim 90\%$ confident” on the correct class, which helps given the known label noise in FER-style datasets.

3.4 Augmentation and regularization

- Geometric/photometric augmentation: horizontal flip, $\pm 15^\circ$ rotation, RandomResizedCrop (0.8–1.0 scale), color jitter (brightness/contrast/saturation), and RandAugment (2 ops, magnitude 9) for additional automated augmentation diversity.

- RandomErasing ($p=0.25$): randomly blanks out a rectangular patch of the image, simulating partial occlusion (e.g., a hand near the mouth), forcing the model to rely on multiple facial regions rather than one.
- Mixup ($\alpha=0.2$), applied to $\approx 50\%$ of training batches: blends two images and their labels in a random ratio λ , with the loss computed as $\lambda \cdot \text{loss}(y_a) + (1-\lambda) \cdot \text{loss}(y_b)$. This smooths the decision boundary between classes and has been shown to particularly help generalization on imbalanced data.

3.5 Optimization

- AdamW optimizer ($lr=3e-4$, $\text{weight_decay}=1e^{-41}$) - weight decay penalizes large weight values, discouraging the model from fitting training-set-specific noise.
- CosineAnnealingLR schedule over 25 epochs — learning rate decays smoothly from $3e-4$ toward 0, allowing large corrective steps early in training and small, precise steps later.
- Automatic Mixed Precision (AMP) via GradScaler/autocast — runs eligible operations in lower precision to speed up training at 224px resolution without a meaningful accuracy cost.

Backpropagation: `loss.backward()` computes the gradient of the loss with respect to every weight via the chain rule; `optimizer.step()` then updates each weight in the direction that reduces that loss. This happens only during training — evaluation and final prediction run in `torch.no_grad()` mode, since no weight updates are needed there.

3.6 Ensemble prediction

Each of the three trained models is reloaded from its best-validation checkpoint (not necessarily its final epoch). For every test image, each model produces two predictions — on the image as-is and on its horizontal mirror (TTA) — which are averaged. The three models' averaged probability vectors are then combined via a validation-accuracy-weighted soft vote (models with higher validation accuracy contribute proportionally more to the final decision), and the class with the highest combined probability is the final prediction.

4. Results and Analysis

4.1 Final submission (Iteration 8)

- Dataset: Train 24,402 / Val 4,307 (stratified 85/15 split, seed 42).
- Best validation accuracy per model: ResNet34 = 0.6835 (epoch 25), EfficientNet-B0 = 0.6845 (epoch 21), ResNet18 = 0.6833 (epoch 25).
- Ensemble weights (proportional to best validation accuracy): ResNet34 = 0.333, EfficientNet-B0 = 0.334, ResNet18 = 0.333 -effectively an equal-weighted vote, since all three models converged to similar validation performance.
- Public leaderboard score: 0.707.
- Training-dynamics note: EfficientNet-B0's train/validation gap widened from -0.05 (epoch 1) to $+0.19$ (epoch 22), the clearest overfitting signal across the three models in this run -consistent with it being the most sensitive to the WeightedRandomSampler's repeated exposure to a small pool of rare-class images.

4.2 Second selected submission (Iteration 7)

- Best validation accuracy: ResNet34 = 0.7042 (epoch 20), EfficientNet-B0 = 0.6963 (epoch 16).
- Ensemble weights: ResNet34 = 0.50, EfficientNet-B0 = 0.50.
- Public leaderboard score: 0.71301

5. Reproducibility

- Environment: Python 3.10, PyTorch + torchvision, timm (ImageNet-pretrained), scikit-learn, Kaggle GPU (T4).
- Fixed random seed (42) and a stratified train/validation split for the final two iterations, so results can be reproduced.
- Only the best-validation-accuracy checkpoint is used for prediction in every iteration from #2 onward, not the final training epoch.
- Full training logs for iterations 7 and 8 (per-epoch validation accuracy, ensemble weights, and console output) are included in the accompanying notebook submission.